

Highlights

- We propose a fusion strategy combining the Mel spectrogram and Gramian Angular Difference Field of transformer voiceprint signals. Gamma correction is applied to enhance the Mel features of the fused image, and a Pulse Coupled Neural Network is employed to improve feature discriminability while suppressing noise interference.
- We design an improved AlexNet-SE architecture within the PINNs framework to achieve accurate fault diagnosis. Specifically, batch normalization and Squeeze-and-Excitation channel attention modules are integrated after each convolutional layer to refine structural representation and enhance channel feature extraction.
- Extensive experiments validate the effectiveness of the proposed method, achieving high recognition accuracy and demonstrating robustness compared with conventional approaches.